

## **Annotation Evaluation Reflection**

### **1) Relevance of Metrics**

The metrics of genome annotation provides important insights into to an organism's genetic characteristics and the overall annotation quality, shown through the numerical values presented. The genome lengths range from 4,726,474 to 4,964,097 base pairs. This range is somewhat small and suggests a possible close relationship between the different bacterial strains, likely belonging to the same or similar species.

The GC (Guanine-Cytosine) content is very consistent, ranging from 52.0% to 52.3%, indicating no signs of contamination or major errors in the sequences, since the numbers are so close to each other. All the genomes coding density falls between 88.4% and 88.9%, indicating the functionality of relevant DNA coding being efficient in use of genome spacing with minimal non-coding regions.

The number of coding sequences (CDS) ranged from 4,336 to 4,611 genes that scale according to genome size, enforcing the assumption that larger genome encode for more proteins. The metric proportions of hypothetical proteins are very informative, since it widely varies from 193 to 486 (4.45%-10.52%). This variation reflections how well the genomes were annotated. The higher percentage genome (GCF\_000007545.1 with a 10.52%) may indicate an incomplete annotation or the lack of genome characterization in the current database. This can also lead to the confidence levels in predictive function being affected. Thus, resulting in limiting the analysis of the pathway reconstruction.

The tRNA and rRNA counts highlights the possibility of inconsistencies within the annotation quality or the assemblies' completeness. The majority of tRNA counts are ranged from 80 to 88, except for a count of 65 (GCF\_000171415.1), this is unusually low. Similarly, most of the rRNA have a count of 22, with two exceptions: one is counted at 2 (GCF\_000171415.1) and another at 33 (GCF\_000171515.1). These unusual counts for tRNA and rRNA could indicate missing, incorrect, or even errors in the annotated assembly regions.

Finally, CRISPR arrays provides further insight into the adaptive immune history of an organism through the presence and numerical value count that ranges from 0 to 6. The difference indicates the presence of an adaptive immune system and that environmental pressures or an event involving horizontal gene transfer occurred. The genome with a count of 6, (GCF\_000171415.1) may suggest that this bacterium had more exposure to foreign DNA, whereas the genome with a 0 (GCF\_000020925.1) may have been relying on other mechanisms for defense or even lost this immune system.

Overall, the provided metrics are crucial in describing the genome structures, assessing the reliability of the annotation, and determining the suitability of the data for comparison and

functionality analyses. While consistent metrics may increase the confidence in the dataset (like GC content and coding density). Outliers potentially highlight areas needing further validation (such as the unusual rRNA, rRNA counts, or even the hypothetical proteins).